Recent Advances In Forensic Anthropology: A Systematic Analysis (2020–2025)

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Abstract

Forensic anthropology has evolved significantly, from traditional skeletal analysis to a highly interdisciplinary science, enriched by advances in imaging, molecular biology and digital technology. The purpose of this scoping review is to examine and synthesise the latest developments in forensic anthropology over the last five years, highlighting key innovations, challenges and future directions. A comprehensive search was made using the databases of PubMed, Medline and Research Gate, using keywords such as forensic anthropology, identification, DNA, and age estimation and a final selection of 41 peer reviewed papers were included.

The review divides the findings into several areas: age and gender estimates, DNA and biomolecular techniques, trauma analysis, virtual anthropology, taphonomy, and ethical considerations. Technologies such as post-mortem CT, 3D imaging, photogrammetry and artificial facial reconstruction are now widely integrated in forensic practice. In addition, the use of microbiomes and orthopaedic implants for identification of difficult cases is being increasingly explored. The ethical and legal frameworks governing the treatment of human remains and digital data have also become more prominent.

Geographically, the majority of contributions come from the US and Europe, with research emerging in Asia. While methodological progress has been remarkable, differences in global standardisation and availability of modern tools persist. This review highlights the need for cross-disciplinary cooperation, training and adherence to ethical standards in order to further strengthen forensic anthropology as a key pillar of the medical-legal system.

Keywords: Forensic Anthropology, Human Identification,3D Imaging, DNA Analysis, Taphonomy, Scoping review.

Introduction

Forensic anthropology is an applied scientific discipline dedicated to identifying unidentified human remains, whether skeletal or in advanced decomposition. It covers a wide range of topics from human physiognomy to osteology and, being multidisciplinary, requires cooperation with other specialized groups. [1]

Since its formal introduction over 80 years ago, forensic anthropology has expanded significantly. Its scope now includes trauma analysis, taphonomy and the increasing involvement of forensic anthropologists in large scale investigations, such as cases of missing or disappeared individuals. Forensic anthropologists also assist in location and recovering remains, often using subdisciplines like forensic archaeology. More recently, practitioners have begun to recognize the importance of ethics in forensic anthropology, emphasizing respectful handling of remains, the professional conduct of experts and the responsible use of technology in the discipline. [2]

Forensic anthropologists utilize traditional techniques to estimate age, sex, race, stature etc using outdated tools like osteometric boards, sliding calipers but now their role extends beyond

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traditional identification by integrating with subdisciplines like forensic archaeology in systematic search, proper retrieval of skeleton, detecting trauma patterns to determine the cause, manner of death and postmortem interval (PMI) estimation by studying human cadaver from crime scene to autopsy suites to reconstruct life history and death circumstances [3-6.]

Forensic anthropology has become a mature scientific field, continuously evolving through innovative, globally-driven research that applies anthropological knowledge to medico-legal problems. With its expanding reach into both established and emerging domains, the demand for high-quality expertise must be supported by continuous education and modernized scientific methods. [3]

Objective

To explore the review of literature on emerging scientific wide array of methods, advancements of anthropology in a forensic perspective during the recent five years methodology

Search strategy

A digital search of Pub Med, Medline and Research Gate was performed along with a manual search of reference lists of relevant articles.

Search/key words

Forensic Anthropology, Recent advances, Forensic, Identification, Wars, Human remains, DNA, Sex, Age, Decomposition

Filters

5 years, Free full text, Articles in English language

Selection Criteria

Articles read in full for the final inclusion and excluded Duplicates, irrelevant articles. The search retrieved 204 results. Of the 204 articles,163 duplicates or irrelevant articles/ eliminated. Results were then refined by their title and abstract. The reference list of all identified articles was further manually searched for additional articles. This process of refining and excluding eventually left a total number of 41 articles. (Fig: 1)

Results

The study was organized into the following key thematic areas.

1. Technological Innovations: The integration of 3D imaging, structured light scanning, photogrammetry and CT has revolutionized the estimation of age and gender. Software tools such as photoshop and blender software are now used for facial similarity reconstruction. Virtual repositories and digital bone collections (e.g. Smithsonian 3D, IMPACT) facilitate global cooperation.
2. DNA and Biomolecular Techniques: The DNA extraction from small or highly degraded samples allows for ancestry and phenotypic analysis. The forensic microbiome is being explored to estimate post-mortem interval and biological sex. Orthopaedic implants, which can be traced through serial numbers, provide valuable leads for identification.
3. Age and Sex Estimation Methods: Acetabular and arthritic changes provide a more accurate estimate of adult age. Ridge density analysis and sinus morphology are increasingly employed for determining gender. PMCT and digital morphometry enhance the precision of stature estimation.
4. Trauma and Pathology Analysis: Advances in pattern recognition of fractures and burn analysis have improved identification in burn

Screening Process for Article Selection:

Identification
Records Identified through data base searching (n=204)

↓

Screening
Records after duplicates removed (n=41)

↓

Eligibility
Full-text articles assessed for eligibility (n=41)

↓

Included
Studies included in qualitative synthesis (n=41)

(Fig: 1) shows flow diagram of screening process

victims. The biomechanics of skull fractures are now digitally modelled for forensic interpretation.

5. Taphonomy and Decomposition Research: Devices such as ARISTA (Netherlands) and those at the University of Tennessee are used for controlled decomposition studies. Remote sensing and minimally invasive techniques enable the detection of grave sites without exhumation.
6. Ethical and Legal Challenges: Global disparities exist in the legal responsibilities for forensic analysis (e.g. the Swiss model supervised by a legal doctor). Ethics guidelines have emerged for the digital facial reconstruction and the respectful handling of virtual remains.
7. Interdisciplinary Collaboration: Multi disciplinary teams involving DNA experts, anthropologists, dentists and pathologists collaborate in the Disaster Victim identification (DVI). The Interpol DVI protocol promotes standardised international practices.
8. Geographic and Temporal Trends: Region-wise, most innovation stems from the U.S., Australia, and Europe, particularly in digital forensics and decomposition research. Asia is contributing population-specific studies, though still limited in volume.
9. Year-wise Analysis: Peak publishing activity occurred between 2019 and 2022, with a noticeable shift from basic anthropometric methods to more advanced biometrics and digital techniques.

Discussion

Age Estimation

Age estimation continues to be pivotal in forensic anthropology, especially in legal contexts such as migration, adoption, child pornography, criminal behavior, and missing persons. Key advances include: 1. Technological Integration: Digital technology like 3D facial prototypes and improved image-processing has enhanced the age estimation (Matthews et al.) [7]. 2. Taphonomic Influences: Studies reported that thermal damage severely affects bone and dental structures and its preservation challenges are critical (Cappella et al. [8] and Adserias Garriga et al. [9]. 3. New Anatomical Markers: Research into acetabulum and its arthritic changes has improved post-mortem age estimation. (Alves-Cardoso & Assis) [10]. 4. Standardization Efforts: The Scientific Working Group for Forensic Anthropology (SWGANTH) [11] emphasizes skeletal system remains as foundational for age estimation protocols.

Sex Estimation

Sex estimation remains a critical first step in biological profiling. Recent developments include: Osteological and Dental Techniques: Traditional methods are still used, but problems remain due to the subjective nature of morphological (non-metric) assessments and the labor-intensive nature of morphometric methods. 3D Imaging advances: Tools such as Intelli Space ® and In Vesalius® enable reproducible and high-resolution assess ments. [12] Dimorphism studies: High accuracy was reported for the pelvis (up to 94%), skull, hyoid, ribs, ulna and metatarsus (up to 94 percent). [13] Fingerprint Ridge Density (FPRD): is gaining because of its population-specific and indivi dualistic properties. [14]

Maxillary Sinus Analysis: CT-based maxillary sinus evaluation provides high accuracy for sex determination, particularly when only fragmentary remains are available. [15]

Stature Estimation:

PMCT offers high resolution imaging and precise measurements without physical handling of the remains, which is an advantage in regions with limited access to skeletal remains. [16]

Decomposition: Decomposition Research Centers in the US, Europe and Australia facilitated semi-controlled trails on post-mortem interval intervals (PMI) under diverse environmental condi tions. [17] ARISTA Facility for Sub-surface Taphonomy and Anthropology employs grave monitoring, remote sensing, and minimally invasive methods to investigate human decomposition. [18] Skeletal Remains Analysis: Historical and modern skeletal remains provide insights into Post-War Identification with improved archival integration and DNA comparison techniques. [19] Burned Remains now leverages heat-induced color gradations (Munsell scale) to reconstruct body positioning and burn dynamics. [20] DNA Recovery: Technological advances now allow DNA profiling of small or degraded bone samples, which helps to

predict ancestry and phenotypic traits. [21]

Virtual Anthropology: Digital advancements have redefined skeletal analysis: 3D Reconstruction: Instruments such as CT, Structured Light Scans (SLS) and laser scanning, generate virtual bone models that optimize the reconstruction of fragmentary remains. Photogrammetry: Cost-effective 3D scanning with DSLR cameras and software such as Agisoft Metashape provides high-resolution virtual bone models[22]. Digital Repositories: Collections like Smithsonian 3D collection and the IMPACT radiologic mummy database offer worldwide access to digitized skeletal samples.

Identification: Facial approximation: Digital reconstruction like eg. Autodesk recap studio Inc™, Blender Foundation™ which determines the steps and the sequence to recreate a face. The soft tissue table was developed by Beaini in a Brazilian for the same region of the skulls for similar purpose. [12] Role of Microbiome: Human DNA can remain unchanged for years and extracts stored at − 20 ◦C or − 80 ◦C can preserve important genetic information for long periods and can be submitted as evidence in court. The composition of the postmortem microbiome differs by biological sex, with Pseudomonas sp. only identified in female cadavers, and Rothia sp. only in male cadavers. The microbial signatures also help in PMI estimation. [23] Craniofacial superimposition: It assist in identification of human skeletal remains when methods of DNA, radiographic comparison and fingerprints cannot be used, relying on photographic overlays rooted in optical physics. [24]

Orthopedic Surgical Implants in Identification: In challenging circumstances when conventional methods of identification like DNA analysis, fingerprints and dental charting are not applicable where visual identity compromised as in dismembered body or skeletal remains in disaster victim identification or excessive thermal alteration cases, Orthopedic surgical implantable devices provide variable and valuable information that helps in identification. [25] Disaster Victim Identification: DVI processes requires collaborative forensic expertise In countries like Switzerland, legal mandates assign DVI duties to forensic pathologists. Interpol DVI, guide standardizes international procedures, ensure appropriate software use, validation and multidisciplinary training. [26]

Osteological Ageing from Digital Data: 3D technologies like laser scanners, CT and micro CT scanners have enabled digital analysis of skeletal remains. Some notable examples of digital collections are the University of Bradford's "From Cemetery to Clinic" and the "Smithsonian's 3D-Digitization" and "Digitized Diseases" and IMPACT Radiological Mummy Database created high quality datasets on the deceased which support forensic casework. (Villa C et al. [10] and Shearer et al. [27]).

Ethics: Ethical dilemmas include issues consent, cultural beliefs and digital replication., As outlined by Marquez-Grant ´ et al [28], questions arise around ownership of data, and presentation of sensitive material. The BABAO guidelines support 3D imaging for educational display, yet concerns regarding equivalency remains. [29] Ultimately, balancing scientific advancement with respect for the deceased and cultural sensitivities is essential to uphold ethical integrity in forensic anthropology.

Conclusion

Forensic Anthropology is undergoing significant advancements which necessitates further studies and research with proper guidelines, in expanding the professional practice of forensic faculty worldwide, for the betterment of the medico-legal system. The multitude of methods tested and the continuous development of new techniques-especially computer aided technologies, high-quality radiological imaging and advances in forensic research fields related to anthropology have improved over the last years and probably continue to do so in the future. Adequate training and funding must be provided to ensure that sound research, based upon the scientific method, remains the cornerstone of advancement in Anthropology specifically in forensic perspective.

Last but not the least, the development of forensic anthropology often depends on a country's legal framework, the volume and typology of forensic casework, and its forensic and educational traditions.

Table 1 : Summary Table

SN	Name Of The 1st Author	Article Name	Type Of Study	Discussed Topic
1	Douglas H. Ubelaker et.al [30]	Recent Advances in Forensic Anthropology	Editorial	Recent Advances
2	Nicholas V. Passalaua et al[31]	The Need to Professionalize Forensic Anthropology	Review Article	Professionalization Of Forensic Anthropology
3	Samiksha Nayyar et.al [32]	Forensic Anthropology: Advancements In Conventional Field in New Millennium	Review Article	Identification
4	Eugénia Cunha et al [33]	Recent Advances in Forensic Anthropological Methods and Research	Editorial	Recent Advances
5	James W.W.Rose et al [34]	Forensic And Expert Social Anthropology as Cultural Expertise: Developing Professional Standards	Paper	Professional Standards
6	Laia Sevillano Oriola et al[35]	Virtual Anthropology: Forensic Applications to Cranial Skeletal Remains from The Spanish Civil War	Original Article	Virtual Anthropology and Cranial Skeletal Remains
7	Abraham Johnson et al[36]	Three-Dimensional Scanning – A Futuristic Technology in Forensic Anthropology	Review Article	Three-Dimensional Scanning
8	Taru Mäkinen et al [8]	The Status and Future of Forensic Archaeology and	Paper	Current And Future Prospects
9	Alfsdotter Clara et al [9]	Development And Implementation Of Forensic	Paper	Ongoing Development Of Forensic
10	Chiara Villa et al [10]	Forensic Anthropology and Archaeology in Denmark	Paper	Status Of Forensic Anthropology in Denmark
11	Purva Wagisha et al [7]	Forensic Anthropology	Chapter	Biological Profiling Of Skeletal Remains
12	Shuai Luo et al[11]	Forensic Age Estimation In Adults By Pubic Bone Mineral Density Using Multidetector Computed Tomography	Original Article	Age Estimation By Md Ct
13	Douglas H. Ubelaker et al[17]	Estimation Of Age in Forensic Anthropology: Historical Perspective and Recent Methodological Advances	Review Article	Recent Research In Age Estimation
14	Cappella A et al [14]	The Issue of Age Estimation in A Modern Skeletal Population: Are Even the More Modern Current Aging Methods Satisfactory for The Elderly	Paper	Age Estimation
15	Adserias-Garriga et al[21]	When Forensic Odontology Met Biochemistry: Multidisciplinary Approach in Forensic Human Identification	Review Article	Age Estimation
16	Alves-Cardoso F[28]	Can Osteophytes Be Used as Age at Death Estimators? Testing Correlations in Skeletonized Human Remains with Known Age-At Death.	Technical Note	Adult Skeleton Biological Profile Estimation
17	Mathews H[29]	Estimating Age and Synthesizing Growth in Children and Adolescents Using 3 D Facial Prototypes	Original Article	Age Estimation By 3d Photographs
18	Forensic Anthropology Subcommittee[20]	Scientific Working Group for Forensic Anthropology (SWGANTH).	Original	Identification
19	Daniel J. Wescott et al[37]	Recent Advances in Forensic Anthropology: Decomposition Research	Review Article	Decomposition Research In Forensic Anthropology
20	Siddharatha Sharma et al[38]	Sex Estimation from Fingerprint Ridge Density. A Review of Literature	Review Article	Sex Estimation

Table 1 : Summary Table

21	Latham Krista E.et al[27]	DNA Recovery and Analysis from Skeletal Material in Modern Forensic Contexts	Review Article	DNA From Skeletal Remains
22	M´Arquez-Grant et al[13]	Ethical Concerns in Forensic Anthropology, In Ethical Approaches to Human Remains	Review Article	Ethics In Forensic Anthropology
23	Rachael M.Carew et al [23]	Drilling Down into Ethics: A Thematic Review of Ethical Considerations for The Creation and Use Of 3D Printed Human Remains in Crime Reconstruction	Review Article	Ethics In 3D Printeed Human Remains
24	Joana Rosa et al[24]	Half A Century of Systematic Research on Heat-Induced Color Changes In Bone – A Review	Review Article	Burned Human Remains
25	Yiran Xu et al[25]	Identifying Human Remains From 20th Century Warfare: A State Of the Field Essay	Review Article	Human Remains Identification
26	Marin Vodanović et al [15]	Artificial Intelligence in Forensic Medicine and Forensic Dentistry	Review Article	Age And Sex Estimation By Artificial Intelligence
27	Hiara Villa et al [16]	Evaluating Osteological Ageing from Digital Data	Review Article	Osteological Aging
28	Laura Maria Beschiu et al[26]	Cranial And Odonatological Met hods for Sex Estimation—A Scoping Review	Review Article	Sex Estimation
29	Giulia Sguazzi et al[39]	Microbial DNA In Human Nucleic Acid Extracts: Recoverability of The Microbiome in DNA Extracts Stored Frozen Long-Term and Its Potential and Ethical	Research Paper	Human DNA From Crime Scene
30	Carl N. Stephan[42]	Craniofacial Superimposition:A Review of Focus Distance Estimation Methods and An Extension to Profile View Photographs	Review Article	Craniofacial Superimposition
31	Abrar Hasan Bukhamseen [18]	The Use of Orthopedic Surgical Devices for Forensic Identification: A Systematic Review	Review Article	Identification
32	Maille Ferreira Nunes Rocha[17]	Sex Estimation by Maxillary Sinus Using Computed Tomography: A Systematic Review	Review Article	Sex Estimation
33	Normaizatul Afizah Ismail[12]	Stature Estimation in The South East Asian Population: A Systematic Review	Review Article	Stature Estimation
34	Christian Gehrig [42]	A Swiss Collaborative Exercise for Disaster Victim Identification (DVI): Covering Situations with Different Levels of Complexity	Research Paper	Disaster Victim Identification
35	Anja Petarosa [18]	Technical Note: The Forensic Anthropology Society of Europe(Fase) Map of Identified Osteological Collections	Technical Note	Osteological Identification
36	Wei Cui [22]	Forensic Characterization and Genetic Portrait of The Gannan Tibetan Ethnic Group Via 165 Ai-Snp Loci	Original Article	Genetic Portrait
37	Ezequiel Ortiz Rosa [43]	Comparative Study of Sex Estimates in Adult Skulls Using Direct Measurement and Tomographic Image Reconstruction	Original Article	Sex Estimation
38	Marine Meucci [44]	The Skeletal Remains of Soldiers from The Two World Wars: Between Identification, Health Research and Memorial Issues	Original Article	Skeletal Remains
39	Julia Gabriela Dietrichkeit Pereira[45]	Applicability Of Forensic Facial Approximation in The Recognition Process of Unclaimed Victims	Original Article	Identification Of Unclaimed Victims
40	Filomena Melchiona [46]	Development And Validation Of Mps-Based System For Human Appearance Prediction In Challenging Forensic Samples	Original Article	DNA Phenotyping
41	RoelofJan Oostraa [47]	Amsterdam Research Initiative For Sub-Surface Taphonomy And Anthropology (Arista) – A Taphonomic Research Facility In The Netherlands For The Study Of Human Remains	Original Article	Human Remains

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